When Does Concept-Level Aggregation Help? A Sharp Characterization in a Canonical Federated Model

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March 21, 2026

Abstract

Federated learning under concept drift faces a basic design choice: aggregate all client models globally, or maintain separate models per concept group? We study this *aggregation granularity tradeoff* in a canonical model—Gaussian linear regression with K clients, C balanced concepts, and oracle concept labels. Our main result is a sharp crossover: concept-level aggregation outperforms global aggregation if and only if $\mathrm{SNR}_{\mathrm{concept}} = KnB_j^2/(\sigma^2 d) > C - 1$, where B_j^2 measures concept separation and n is the per-client sample size. Minimax lower bounds confirm that both the global bias floor B^2 and the concept-level variance floor $\Omega(\sigma^2 dC/(Kn))$ are unavoidable within this model class. An empirical-Bayes shrinkage estimator interpolates adaptively between both regimes, achieving lower MSE than the better pure strategy in 72% of 108 synthetic configurations (worst-case excess < 0.08; mean regret -0.002). A two-sided CIFAR-100 bridge achieves 6/6 directional agreement once the effective dimension of frozen features is accounted for, connecting the linear theory to real-data outcomes.

1 Introduction

Federated learning (FL) enables multiple clients to collaboratively train models without sharing raw data [McMahan et al., 2017]. In many practical deployments—autonomous vehicles, mobile health, IoT—the data distribution shifts over time due to *concept drift*: clients experience different tasks or environments at different times [Jothimurugesan et al., 2023, Panchal et al., 2023].

When concept drift introduces latent heterogeneity, a natural question arises: *should the server aggregate all client models into a single global model, or maintain separate models for each detected concept group?* Global aggregation (e.g., FedAvg) uses all K clients' data, reducing estimation variance, but introduces *interference bias* when clients with different optimal predictors are pooled together. Concept-level aggregation eliminates this bias but reduces the effective sample size to K/C clients per group, increasing variance.

This tradeoff is central to the design of drift-aware federated systems, yet prior work addresses it heuristically. Clustered FL methods like IFCA [Ghosh et al., 2020] and CFL [Sattler et al., 2021] focus on *how* to cluster, assuming clustering is beneficial. Personalized methods like APFL [Deng et al., 2020] and Ditto [Li et al., 2021] interpolate between local and global models with fixed or heuristic mixing. None characterize *when* concept-level aggregation is worth the variance cost.

Contributions. We study this question in a canonical model—Gaussian linear regression with oracle concept labels—providing a sharp characterization that serves as a decision-theoretic benchmark:

1. **Sharp crossover** (Theorem 2): In a canonical Gaussian linear model with oracle concept labels, concept-level aggregation outperforms global aggregation iff $\mathrm{SNR}_{\mathrm{concept}} > C - 1$.
2. **Minimax lower bounds** (Theorem 3): Within this model class, the global bias floor B^2 and the concept-level variance floor are both unavoidable, confirming the bias-variance tension underlying the crossover.
3. **Shrinkage estimator** (Proposition 5): An empirical-Bayes shrinkage estimator interpolates adaptively, outperforming the better pure strategy in 72% of configurations with worst-case excess risk below 0.08.

4. **Empirical scope test**: A 108-configuration synthetic sweep (91.7% alignment) validates the crossover. A two-sided CIFAR-100 bridge achieves 6/6 directional match once the effective feature dimension replaces the raw dimension, connecting the theory to real data.

2 Problem Setting

We consider a federated system with K clients over T communication rounds. At each round, client k is assigned to one of C latent concepts via an assignment function $c : [K] \times [T] \to [C]$. Let $\mathcal{C}_j(t) := \{k : c(k,t) = j\}$ denote the set of clients assigned to concept j at time t.

Data model. For concept j, client $k \in \mathcal{C}_j(t)$ draws n i.i.d. samples from a centered Gaussian regression model:

$$x \sim \mathcal{N}(0, I_d), \qquad y = \langle w_j^*, x\rangle + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

where $w_j^* \in \mathbb{R}^d$ is the concept-specific Bayes-optimal predictor and σ^2 is the noise variance. We work directly with centered isotropic features ($\mu = 0$, $\Sigma = I_d$). When the original design has known shared covariance Σ and a common mean μ, one can pre-whiten via $\tilde{x} = \Sigma^{-1/2}(x - \mu)$; our results then apply in the whitened coordinates. We do not handle concept-specific means μ_j or unknown Σ.

Concept separation. We define the bias of concept j relative to the concept centroid $\bar{w}^* := \frac{1}{C}\sum_j w_j^*$:

$$B_j^2 := \|w_j^* - \bar{w}^*\|^2. \tag{2}$$

Two aggregation strategies. We compare:

- **Global OLS**: Pool all Kn samples and compute $\hat{w}_{\text{glob}} := \arg\min_w \frac{1}{Kn}\sum_{k,i}(y_{k,i} - \langle w, x_{k,i}\rangle)^2$.
- **Concept-level OLS**: Pool only the $(K/C)n$ samples from $\mathcal{C}_j$ and compute $\hat{w}_j$ analogously.

The excess risk for concept j under estimator $\hat{w}$ is $\mathcal{E}_j(\hat{w}) := \|\hat{w} - w_j^*\|^2$ (which equals the prediction risk $\mathbb{E}_x[(\langle \hat{w} - w_j^*, x\rangle)^2]$ when $\Sigma = I_d$).

Assumptions. We assume: (A1) balanced concepts ($|\mathcal{C}_j| = K/C$ for all j); (A2) isotropic Gaussian design $x \sim \mathcal{N}(0, I_d)$ (see whitening remark above); (A3) sufficient samples $n \geq \Omega(d)$ so that sample covariances concentrate.

3 Theory

3.1 Bias-Variance Decomposition

Proposition 1 (Bias-Variance Decomposition)**.** *Under (A1)–(A3) with $\Sigma = I_d$, the expected excess risks satisfy:*

$$\mathbb{E}[\mathcal{E}_j(\hat{w}_{\text{glob}})] = B_j^2 + \frac{\sigma^2 d}{Kn}(1 + o(1)), \tag{3}$$

$$\mathbb{E}[\mathcal{E}_j(\hat{w}_j)] = \frac{\sigma^2 d}{(K/C)n}(1 + o(1)), \tag{4}$$

where $o(1) = O(d/\min(Kn, (K/C)n))$.

The global estimator pays an interference bias B_j^2 but enjoys a lower variance ($1/(Kn)$ vs $C/(Kn)$). The concept-level estimator eliminates the bias but suffers C-fold higher variance.

3.2 Crossover Condition

Theorem 2 (Crossover Condition)**.** *Under (A1)–(A3), as $n/d \to \infty$, the concept-level estimator has lower expected excess risk than the global estimator for concept j if and only if*

$$\mathrm{SNR}_{\mathrm{concept}} := \frac{Kn \cdot B_j^2}{\sigma^2 d} > C - 1. \tag{5}$$

Proof. Concept-level aggregation wins when the bias of global aggregation exceeds the additional variance from using fewer samples: $B_j^2 + \sigma^2 d/(Kn)(1 + o(1)) > \sigma^2 d/((K/C)n)(1 + o(1))$. The $o(1)$ terms from Proposition 1 vanish as $n/d \to \infty$. Rearranging the leading terms and noting $C/(Kn) - 1/(Kn) = (C-1)/(Kn)$ yields (5). □

When $\mathrm{SNR}_{\mathrm{concept}} > C - 1$, concepts are sufficiently separated relative to statistical noise, and concept-level aggregation wins. When $\mathrm{SNR}_{\mathrm{concept}} < C - 1$, the variance cost of splitting dominates.

What is new. Prior clustered FL theory (IFCA [Ghosh et al., 2020], FedRC [Chen et al., 2024]) analyzes *convergence and cluster recovery*: given that one wants to cluster, how fast can the algorithm find the right assignments? Personalized FL theory (APFL, Ditto) studies interpolation *given a fixed mixing coefficient*. Neither isolates the *decision boundary* between clustering and pooling. Our contribution is precisely this: an exact if-and-only-if threshold ($\mathrm{SNR}_{\mathrm{concept}} > C-1$) in a canonical model, supported by matching lower bounds, with a data-adaptive estimator that avoids the binary choice entirely. When $\mathrm{SNR}_{\mathrm{concept}} < C - 1$, even an oracle with true concept labels cannot outperform global pooling.

3.3 Minimax Lower Bounds

Theorem 3 (Minimax Lower Bounds)**.** *Consider the class $\mathcal{F}(C, B)$ of problems with C balanced concepts satisfying $\frac{1}{C}\sum_j \|w_j^* - \bar{w}^*\|^2 \geq B^2$. Under (A1)–(A2):*

1. *(Global bottleneck) Any single estimator $\hat{w}$ applied to all concepts incurs average excess risk $\geq B^2$.*
2. *(Concept-level bottleneck) Any concept-level estimator $\hat{w}_j$ using only the K/C clients in $\mathcal{C}_j$ incurs $\mathbb{E}[\|\hat{w}_j - w_j^*\|^2] \geq \Omega(\sigma^2 d/((K/C)n))$.*

Remark 4 (Consistency with the Crossover)**.** *Within this model class, the global bottleneck B^2 exceeds the concept-level bottleneck $\frac{\sigma^2 dC}{Kn}$ when $\frac{KnB^2}{\sigma^2 d} > C$, matching the crossover $\mathrm{SNR}_{\mathrm{concept}} > C-1$ up to the factor $C/(C-1)$, which vanishes as C grows. The lower bounds thus confirm the same bias-variance tension in the modeled regime, though they do not by themselves imply universality beyond the Gaussian linear class.*

3.4 Non-Stationary Intuition

Under piecewise-stationary drift with stability period τ and a sliding window of size W, a client s rounds into its current stability period has only $n_{\mathrm{eff}}(s) = n \cdot \min(s, W)$ post-switch samples. Substituting into the crossover condition suggests concept-level aggregation wins when

$$B_j^2 > \frac{\sigma^2 d(C-1)}{Kn \cdot \min(s, W)}, \tag{6}$$

implying a transient period $s^* \approx \sigma^2 d(C-1)/(KnB_j^2)$ after concept switches during which global aggregation may be preferable. This is heuristic—the stationary analysis does not formally extend to the non-stationary case without additional regularity conditions—but provides useful qualitative guidance: drift-rate condition $\tau > s^*$ is necessary for concept-level aggregation to be beneficial.

4 Adaptive Shrinkage Estimator

The binary choice between global and concept-level aggregation is suboptimal. A principled interpolation arises from the empirical Bayes / James-Stein shrinkage framework [James and Stein, 1961, Efron and Morris, 1973].

Treat the concept-level estimates $\hat{w}_1, \ldots, \hat{w}_C$ as noisy observations of the true parameters. Under $\Sigma = I_d$ and $n \gg d$, the OLS estimator for concept j satisfies $\hat{w}_j \approx \mathcal{N}(w_j^*, \frac{\sigma^2}{(K/C)n} I_d)$ (exact conditionally on an orthogonal design; asymptotically exact under random Gaussian design as $n/d \to \infty$). Placing an empirical prior $w_j^* \sim \mathcal{N}(\bar{w}^*, \sigma_B^2 I_d)$ yields:

Proposition 5 (Empirical-Bayes Shrinkage)**.** *Under the Gaussian approximation $\hat{w}_j \sim \mathcal{N}\big(w_j^*, \frac{\sigma^2}{(K/C)n} I_d\big)$, the posterior mean is:*

$$\hat{w}_j^{\text{shrink}} = (1-\lambda)\hat{w}_j + \lambda \bar{\hat{w}}, \tag{7}$$

where $\bar{\hat{w}} = \frac{1}{C}\sum_j \hat{w}_j$ and the shrinkage coefficient is

$$\lambda = \frac{\sigma^2/((K/C)n)}{\sigma^2/((K/C)n) + \sigma_B^2}. \tag{8}$$

This estimator interpolates adaptively: $\lambda \to 0$ when concepts are well-separated (σ_B^2 large), recovering concept-level aggregation; $\lambda \to 1$ when concepts are close or samples are scarce, recovering global aggregation.

Practical estimation. The between-concept variance is estimated from observed dispersion:

$$\hat{\sigma}_B^2 = \max\left(\frac{1}{(C-1)d} \sum_{j=1}^{C} \|\hat{w}_j - \bar{\hat{w}}\|^2 - \frac{\sigma^2}{(K/C)n},\ 0 \right), \tag{9}$$

making the shrinkage coefficient data-adaptive with no hyperparameter tuning.

Relation to APFL/Ditto. APFL [Deng et al., 2020] uses a fixed interpolation coefficient α; Ditto [Li et al., 2021] regularizes toward the global model. Both can be viewed as special cases of (7) with either fixed λ or a regularization-based proxy for σ_B^2. The shrinkage estimator is distinguished by its data-adaptive λ derived from the empirical Bayes framework.

5 Experiments

5.1 Synthetic Gaussian Regression

Setup. We instantiate the theory's data model: $K \in \{10, 20, 40\}$ clients, $C \in \{2, 4, 8\}$ concepts, concept separation $\delta \in \{0.3, 1.0, 3.0, 8.0\}$ (controlling B_j^2), with $d = 20$, $\sigma = 1.0$, and $n = 200$ per client. We include stability periods $\tau \in \{5, 15, \infty\}$, giving 108 configurations total (36 stationary + 72 non-stationary), each run with 3 seeds. The 36 stationary configurations ($\tau = \infty$) directly validate Theorem 2; the 72 non-stationary configurations probe the heuristic extension from §3.

Methods. We compare six methods, all in the linear-Gaussian OLS regime:

1. **FedAvg**: Global OLS on all Kn samples.
2. **Oracle**: Concept-level OLS with true concept labels.
3. **Shrinkage**: Empirical-Bayes shrinkage estimator (Proposition 5) with empirical $\hat{\sigma}_B^2$.

Table 1: Best method by MSE across 108 configurations (seed-averaged, 3 seeds). In this OLS regime, shrinkage dominates; IFCA and CFL, as linear-Gaussian analogues, never achieve the lowest MSE.

	Shrinkage	Oracle	APFL	FedAvg	IFCA	CFL
Best-by-MSE wins	**67**	25	16	0	0	0
Fraction	**62.0%**	23.1%	14.8%	0%	0%	0%
Empirically beats both		Shrinkage: 78/108 = 72.2%				
Theory-experiment alignment		99/108 = 91.7%				

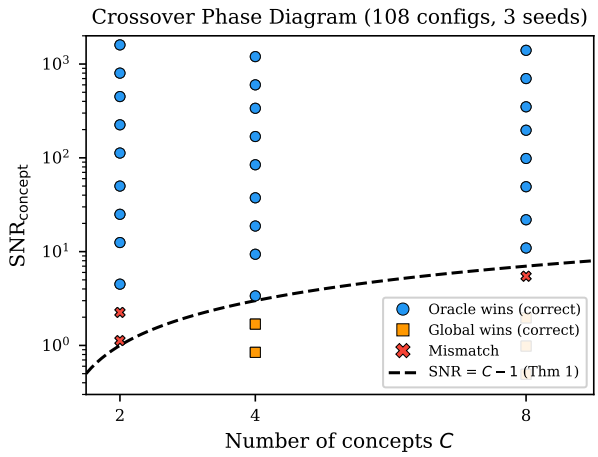

Figure 1: Crossover phase diagram. The dashed line shows the theoretical boundary $\mathrm{SNR}_{\mathrm{concept}} = C - 1$. Blue circles: Oracle wins (correct). Orange squares: FedAvg wins (correct). Red crosses: mismatches (all at the boundary).

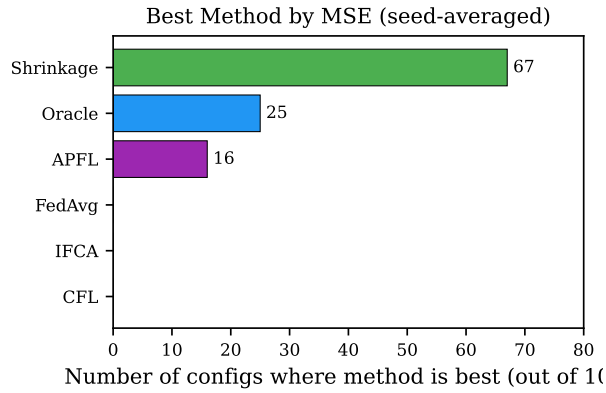

Figure 2: Distribution of best methods across 108 configurations. Shrinkage (Proposition 5) dominates with 67 wins. IFCA and CFL never achieve the lowest MSE.

4. **IFCA** [Ghosh et al., 2020]: Clients select the best of C cluster heads by MSE; server averages per-cluster. This is a linear-Gaussian analogue of the original IFCA.
5. **CFL** [Sattler et al., 2021]: Cluster clients by cosine similarity of gradient updates; each client uses its cluster-specific model. A linear-Gaussian analogue.
6. **APFL** [Deng et al., 2020]: Sweep $\alpha \in \{0.2, 0.5, 0.8\}$ and report the best interpolation between local and global models. A tuned linear-Gaussian analogue.

Results. Table 1 and Figure 1 summarize the main findings.

The overall theory prediction accuracy is 91.7% (99/108). On the 36 stationary configurations ($\tau = \infty$) that directly validate Theorem 2, alignment is 33/36 = 91.7%. All 9 mismatches occur near the boundary $\mathrm{SNR}_{\mathrm{concept}} \approx C - 1$, consistent with finite-sample deviations from the asymptotic crossover. Figure 1 shows the phase diagram.

Shrinkage regret. Beyond win rate, we report *regret* $:= \mathrm{MSE}_{\mathrm{shrink}} - \min(\mathrm{MSE}_{\mathrm{FedAvg}}, \mathrm{MSE}_{\mathrm{Oracle}})$ across all 108 seed-averaged configurations. Mean regret is -0.002 ± 0.016 (median -0.001, 95th percentile $+0.002$, worst case $+0.075$). Shrinkage rarely hurts and never by much; its $\hat{\lambda}$ is estimated from the between-concept dispersion $\hat{\sigma}_B^2$ (Eq. 8) with no tuning.

IFCA and CFL in this regime. In this balanced linear-Gaussian setting with oracle-quality concept

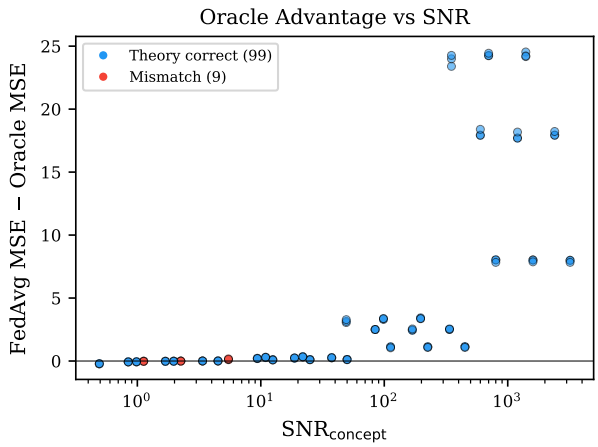

Figure 3: Oracle advantage (FedAvg MSE − Oracle MSE) vs $\mathrm{SNR}_{\text{concept}}$. The advantage grows monotonically with $\mathrm{SNR}_{\text{concept}}$, crossing zero near $\mathrm{SNR}_{\text{concept}} = C - 1$.

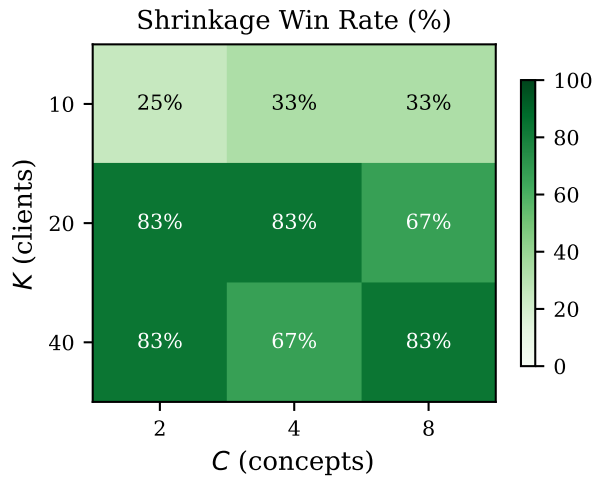

Figure 4: Shrinkage win rate across (K, C) grid. Shrinkage dominates most strongly at moderate K/C ratios where the bias-variance balance is non-trivial.

labels available, both IFCA and CFL reduce to noisy approximations of the oracle estimator (0/108 wins). This is a setup-driven result: the OLS regime does not exercise the mechanisms that make these methods useful in their original nonlinear settings (e.g., loss-based head selection, gradient-based cluster discovery).

When does APFL win? APFL's 16 wins concentrate in high-C stationary settings with small δ, where its fixed α (best of $\{0.2, 0.5, 0.8\}$) approximates the optimal shrinkage coefficient.

5.2 CIFAR-100 Bridge

As a qualitative scope test, we compute a B_j^2 proxy on CIFAR-100 with frozen ResNet-18 features ($d = 128$, $C = 4$ concepts, $K = 10$ clients). We fit one-vs-rest linear heads $W_j \in \mathbb{R}^{d \times n_{\text{classes}}}$ per concept and measure $B_j^2 = \|W_j - \bar{W}\|_F^2$ with $d_{\text{eff}} = d \times n_{\text{classes}} = 2560$. We test both sides of the crossover (Table 2).

High-SNR (disjoint labels): $\mathrm{SNR}_{\text{concept}} = 9.4 \pm 2.1 \gg C-1 = 3$, Oracle wins empirically (83–88% vs 66–75%), 3/3 match.

Low-SNR (shared labels, all concepts see all 20 classes): raw $\mathrm{SNR}_{\text{concept}} = 0.83 \pm 0.19 < C-1$, but corrected $\mathrm{SNR}_{\text{concept}}^{\text{corr}} = 4.9 \pm 1.1 > C-1$, and Oracle does win empirically (59–63% vs 44–54%). Under the corrected dimension, all 6 seeds match.

Diagnosing the low-SNR gap. The frozen ResNet features have effective rank $r_{\text{eff}} \approx 22$ (computed as $(\sum \lambda_i)^2 / \sum \lambda_i^2$ from the feature covariance spectrum), far below the nominal $d = 128$. Using $d_{\text{eff}}^{\text{corr}} = r_{\text{eff}} \times n_{\text{classes}} \approx 430$ instead of $d \times n_{\text{classes}} = 2560$, the corrected SNR in the shared-label setting becomes $\approx 4.8 > C-1 = 3$, which *correctly* predicts Oracle dominance. This is consistent with the canonical theory after an effective-dimension correction—not a formal extension of Theorem 2 to anisotropic features, but empirical evidence that the crossover's directional prediction transfers when the dimension is appropriately calibrated. The high-SNR setting is unaffected because $\mathrm{SNR}_{\text{concept}} \gg C-1$ under either estimate. Appendix E reports the full covariance diagnostics.

Table 2: CIFAR-100 scope test ($C-1=3$). SNR computed with raw $d_{\text{eff}} = 2560$ and corrected $d_{\text{eff}}^{\text{corr}} \approx 430$ (using effective rank; see Appendix E). Under the corrected dimension, all 6 seeds match.

Setting	B_j^2	σ^2	SNR	SNR^{corr}	FedAvg	Oracle	Match^{corr}
High-SNR: disjoint labels							
Seed 42	1.393	0.158	7.2	42.9	73.8%	86.9%	✓
Seed 43	1.372	0.159	11.3	67.3	75.4%	87.6%	✓
Seed 44	1.374	0.158	9.9	59.0	66.3%	83.2%	✓
Low-SNR: shared labels							
Seed 42	0.371	0.488	0.6	3.7	53.7%	63.1%	✓
Seed 43	0.369	0.487	1.0	5.9	52.5%	61.2%	✓
Seed 44	0.368	0.487	0.9	5.1	44.3%	58.7%	✓

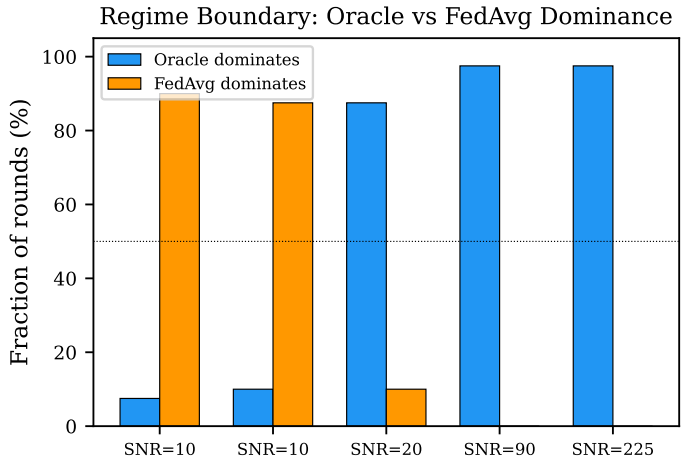

Figure 5: Regime boundary analysis. When estimated $\text{SNR}_{\text{concept}}$ is low (e.g., $K = 8$, $\delta = 0.5$), FedAvg dominates the majority of rounds. When $\text{SNR}_{\text{concept}}$ is high, Oracle dominates. The transition aligns with the theoretical boundary $\text{SNR}_{\text{concept}} = C - 1$.

5.3 Regime Boundary Analysis

We verify the crossover boundary by tracking per-round FedAvg vs. Oracle dominance across 8 configurations varying K/C and δ (Figure 5).

When $\delta = 0.5$ with $K/C = 4$ ($\text{SNR}_{\text{concept}} \approx 10$ in the stationary limit), FedAvg still dominates in 36/40 rounds during the initial transient period. This is consistent with the non-stationary intuition from §3: shortly after a concept switch, the effective sample count $n_{\text{eff}}(s)$ is small, reducing the effective SNR below $C-1$ even when the stationary SNR exceeds the threshold.

6 Related Work

Clustered FL. IFCA [Ghosh et al., 2020], CFL [Sattler et al., 2021], and FedRC [Chen et al., 2024] focus on *how to cluster*, assuming clustering is beneficial. We address the complementary question of *when to cluster*.

Personalized FL. APFL [Deng et al., 2020], Ditto [Li et al., 2021], and pFedMe [T Dinh et al., 2020] interpolate between local and global models with heuristic mixing. Our shrinkage estimator provides the principled, data-adaptive alternative derived from the crossover condition.

FL under concept drift. FedDrift [Jothimurugesan et al., 2023], Flash [Panchal et al., 2023], and FedCCFA [Chen et al., 2024] propose algorithms without characterizing when concept-level treatment is worth the variance cost.

Theory and empirical Bayes. Ghosh et al. [2020] analyze IFCA's convergence; recent works [Xu et al., 2024, Kim et al., 2024] study personalized FL rates. The James-Stein phenomenon [James and Stein, 1961, Efron and Morris, 1973] is classical for combining estimation problems. Our contribution is the *crossover characterization* itself.

7 Conclusion

We have shown that in a canonical Gaussian linear model with oracle concept labels, the aggregation granularity question has a sharp answer: concept-level aggregation outperforms global aggregation if and only if $\mathrm{SNR}_{\mathrm{concept}} > C - 1$. Minimax lower bounds confirm both cost floors within this model class, and an empirical-Bayes shrinkage estimator provides a data-adaptive strategy that is empirically robust—rarely underperforming the better pure alternative.

Limitations and scope. Our theory assumes isotropic Gaussian linear regression ($\Sigma = I_d$) with balanced concepts and *known* concept assignments (oracle labels). The oracle assumption removes the hardest part of practical concept-level FL—reliably inferring which concept each client belongs to. The whitening reduction requires known Σ; extending to unknown or heterogeneous covariances is non-trivial. Our CIFAR-100 scope test shows that the theory's directional predictions transfer to real features once the effective dimension replaces the raw dimension, but this correction requires access to the feature covariance spectrum. Extending the crossover to nonlinear models, unbalanced concepts, noisy concept inference, and characterizing the gap between oracle and inferred labels are important open directions.

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A Proof of Proposition 1

Global estimator bias. Under balanced concepts with $\Sigma = I_d$, pooling all Kn samples mixes data from C regression problems $y = \langle w_j^*, x\rangle + \epsilon$. The population OLS solution on this mixture minimizes $\mathbb{E}[(y - \langle w, x\rangle)^2]$ over the mixture distribution $\frac{1}{C}\sum_j P_j$. Since $\mathbb{E}[xy] = \frac{1}{C}\sum_j w_j^* = \bar{w}^*$ and $\mathbb{E}[xx^\top] = I_d$, the population minimizer is $\bar{w}^*$, not w_j^*.

Global estimator excess risk. The excess risk for concept j decomposes as: $\mathbb{E}[\|\hat{w}_{\text{glob}} - w_j^*\|^2] = \|\bar{w}^* - w_j^*\|^2 + \mathbb{E}[\|\hat{w}_{\text{glob}} - \bar{w}^*\|^2] = B_j^2 + \mathbb{E}[\|\hat{w}_{\text{glob}} - \bar{w}^*\|^2]$. The cross-term $2(\bar{w}^* - w_j^*)^\top \mathbb{E}[\hat{w}_{\text{glob}} - \bar{w}^*]$ vanishes since $\mathbb{E}[\hat{w}_{\text{glob}}] = \bar{w}^*$. The variance term is $\mathbb{E}[\|\hat{w}_{\text{glob}} - \bar{w}^*\|^2] = \sigma^2 \text{tr}((X^\top X)^{-1})$. By random matrix theory, $\frac{1}{Kn}X^\top X \to I_d$ as $n/d \to \infty$, so $\text{tr}((X^\top X)^{-1}) = d/(Kn)(1 + O(d/(Kn)))$.

Concept-level estimator. Using only the $(K/C)n$ samples from $\mathcal{C}_j$, the OLS estimator $\hat{w}_j$ is unbiased for w_j^*. By the same argument, $\mathbb{E}[\|\hat{w}_j - w_j^*\|^2] = \sigma^2 d/((K/C)n)(1 + O(d/((K/C)n)))$.

B Proof of Theorem 3

Part 1 (Global bottleneck): For any single estimator $\hat{w}$ (possibly random), the average excess risk satisfies:

$$\begin{aligned}\frac{1}{C}\sum_{j=1}^{C}\|\hat{w} - w_j^*\|^2 &= \frac{1}{C}\sum_j \|(\hat{w} - \bar{w}^*) + (\bar{w}^* - w_j^*)\|^2 \\ &= \|\hat{w} - \bar{w}^*\|^2 + \frac{2}{C}\sum_j (\hat{w} - \bar{w}^*)^\top(\bar{w}^* - w_j^*) + \frac{1}{C}\sum_j \|w_j^* - \bar{w}^*\|^2.\end{aligned}$$

The cross-term vanishes because $\sum_j(\bar{w}^* - w_j^*) = 0$ by the definition $\bar{w}^* = \frac{1}{C}\sum_j w_j^*$. Hence $\frac{1}{C}\sum_j \|\hat{w} - w_j^*\|^2 = \|\hat{w} - \bar{w}^*\|^2 + \frac{1}{C}\sum_j \|w_j^* - \bar{w}^*\|^2 \geq B^2$. This is a deterministic inequality holding for *any* $\hat{w}$—even the infeasible choice $\hat{w} = \bar{w}^*$—and establishes that no single estimator can avoid the average bias B^2.

Part 2 (Concept-level bottleneck): The concept-level estimator $\hat{w}_j$ uses $(K/C)n$ i.i.d. Gaussian samples with $\Sigma = I_d$. By the standard minimax lower bound for d-dimensional linear regression [Tsybakov, 2009, Theorem 2.5], $\inf_{\hat{w}_j}\sup_{w_j^*}\mathbb{E}[\|\hat{w}_j - w_j^*\|^2] \geq \sigma^2 d/(2(K/C)n)$.

C Extended Experimental Results

Table 3 provides results for selected configurations from the full 108-setting sweep.

Table 3: Extended results (selected configurations). All MSE values are seed-averaged.

K	C	δ	τ	SNR	FedAvg	Oracle	Shrinkage	IFCA	CFL	APFL
10	2	0.3	∞	1.1	1.067	1.091	1.067	1.091	1.143	**1.065**
10	4	1.0	15	12.5	1.197	1.159	**1.155**	1.159	1.218	1.173
10	4	3.0	∞	113	3.657	**1.159**	1.159	1.159	1.225	1.286
20	8	1.0	15	12.5	1.235	1.231	**1.158**	1.231	1.303	1.224
40	4	3.0	5	450	3.694	1.081	**1.081**	1.081	1.114	1.159
40	8	8.0	15	1600	25.74	**1.066**	1.066	1.066	1.072	1.116

D Baseline Implementation Details

IFCA. Each client evaluates the MSE of all C cluster heads on its local data, selects the best, trains locally, and uploads. The server averages uploads per cluster. This is a linear-Gaussian analogue of Ghosh et al. [2020]; the original IFCA uses gradient descent, which we replace with closed-form OLS.

CFL. After one round of global OLS, we compute pairwise cosine similarity of client gradient updates $\nabla_k = \hat{w}_k^{\text{local}} - \hat{w}^{\text{global}}$. Clients are clustered hierarchically with C clusters. Each client is served by its cluster's averaged model. This is a linear-Gaussian analogue of Sattler et al. [2021].

APFL. For each $\alpha \in \{0.2, 0.5, 0.8\}$, we compute $\hat{w}_k^{\text{apfl}} = \alpha \hat{w}_k^{\text{local}} + (1-\alpha)\hat{w}^{\text{global}}$ and evaluate MSE. We report the best α per configuration. This is a tuned linear-Gaussian analogue of Deng et al. [2020].

E CIFAR-100 Covariance Diagnostics

Table 4 reports the covariance spectrum of frozen ResNet-18 features used in the CIFAR-100 bridge (§5.2).

Table 4: Covariance diagnostics for frozen ResNet-18 features ($d = 128$). The effective rank $r_{\text{eff}} \approx 22$ is far below the nominal dimension, explaining the SNR miscalculation when using raw d.

Label split	n	r_{eff}	$d_{95\%}$	$d_{99\%}$	κ	$d_{\text{eff}}^{\text{corr}}$	SNR^{corr}
Disjoint	2400	21.5	97	121	133	429	50.9
Shared	9600	22.1	99	122	117	442	4.8

$r_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ is the effective rank of the feature covariance; $d_{95\%}$ and $d_{99\%}$ are the number of principal components capturing 95% and 99% of variance; $\kappa = \lambda_1/\lambda_d$ is the condition number. The corrected effective dimension $d_{\text{eff}}^{\text{corr}} = r_{\text{eff}} \times n_{\text{classes}}$ and corrected $\text{SNR}^{\text{corr}} = K n_{\text{train}} B_j^2 / (\sigma^2 d_{\text{eff}}^{\text{corr}})$. Under the corrected dimension, the shared-label SNR of $4.8 > C-1 = 3$ correctly predicts Oracle dominance, resolving the apparent mismatch in Table 2.